MIL-W-19487A(OS)
24 June 1975
SUPERSEDING
MIL-W-19487(NORD)
14 May 1956
and Amendment 1
6 September 1957

MILITARY SPECIFICATION

WIRE, NICKEL-BASE ALLOY, AGE HARDENABLE

This specification is approved for use by all departments and agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers two grades of nickel-base alloy wire for coil spring applications requiring high strength and resistance to corrosion.

1.2 Classification. Nickel-base alloy wire shall be of the following grades, and conditions, as specified (see 6.2).

Grade A - Nickel-aluminum-titanium alloy
Grade B - Nickel-titanium-magnesium alloy

Condition 1 - Spring temper, as-drawn

a. Grade A wire (see 1.3.2) which has been given a final reduction of 60 to 70 percent after the final process anneal at 1600° to 1800° Fahrenheit (F).

b. Grade B wire (see 1.3.2) which has been given a final reduction of 60 to 70 percent after the final process anneal. The final process anneal shall consist of a treatment response to subsequent age hardening; where practical and for large sizes, water quenching shall be used.

Condition 2 - Spring temper, age-hardened

a. Grade A or grade B wire which has been processed as desired for condition 1 and subsequently aged (see note 6.3).

FSC 9525

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:54Z
Check the source to verify that this is the current version before use.

1.3 Fabrication.

1.3.1 Unless otherwise specified in the contract or order, nickel-base alloy wire shall be furnished in condition 1 suitable for coiling and subsequent age hardening by the fabricator.

1.3.2 Nickel-base alloy wire shall be cold-drawn from hot finished wire or rod which has been previously ground or processed for the removal of seams, cracks, or other injurious surface defects.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein. In the event of conflict between this specification and other documents referenced herein, requirements of this specification shall apply.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Table for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-163	Steel Mill Products, Preparation for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 The material used shall be such as to produce nickel-base alloy wire in full compliance with the chemical and mechanical requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:54Z
Check the source to verify that this is the current version before use.

3.1.2 Macroscopic. The wire shall be uniform in quality and condition. Visual examination of samples after deep acid etch (see 4.3.3) shall disclose the wire to be clean, sound, smooth, and free from foreign material, pipes, slivers, laps, cracks, welds, twists, kinks, seams, scale, damaged ends or edges, buckles, or other defects injurious to the normal performance of a spring.

3.2 Chemical composition. Chemical composition of the material is given in table I (see 6.4). An analysis of each lot of wire shall be furnished by the contractor showing the percentages of each element designated.

Table I

CHEMICAL COMPOSITION (%)

Elements	Grade A	Grade B
Nickel,[1] minimum	93.0	97.0
Aluminum	4.0 - 4.75	-
Carbon (total), maximum	0.30	0.40
Iron, maximum	0.60	0.60
Manganese, maximum	0.50	0.50
Silicon, maximum	1.0	0.35
Titanium	0.25 - 1.00	0.20 - 0.60
Sulfur, maximum	0.010	0.010
Copper, maximum	0.25	0.25
Magnesium	-	0.20 - 0.50

[1] Nickel including cobalt.

3.3 Mechanical properties. Mechanical properties shall be as shown in table II.

3.3.1 Response to ageing treatment. Grades A and B material ordered in the unaged condition (condition 1) shall be demonstrated to be capable of meeting the mechanical properties of the corresponding heattreated condition (condition 2) shown in table II.

3.4 Wrapping.

3.4.1 Condition 1. Wire of condition 1 up to 0.229 inch diameter, inclusive, shall withstand wrapping, eight consecutive close wound turns,

Source: https://assist.dla.mil -- Downloaded: 202610-18T12:54Z
Check the source to verify that this is the current version before use.

without cracking, around a rod of the same diameter. Wire over 0.229 inch in diameter shall withstand a similar wrapping, without cracking, around a rod of twice the wire diameter.

Table II

TENSILE STRENGTH OF COLD-DRAWN WIRE IN COILS, GRADES A AND B

Condition and size (inch)	Tensile strength[1] (psi, minimum)
Condition 1	
Cold-drawn, spring, spring temper (as-drawn)	
0.057 and less	170,000
Over 0.057 to 0.229, inclusive	160,000
Over 0.229 to 0.312, inclusive	155,000
Condition 2	
Cold-drawn, spring temper and age-hardening	
0.057 and less	205,000
Over 0.057 to 0.229, inclusive	200,000
Over 0.229 to 0.312, inclusive	190,000

[1]Applicable to material in coil. For material in straightened and cut lengths, deduct 15,000 psi from above values in both the as-drawn and as-drawn and age-hardened material.

3.4.2 Condition 2. Wire of condition 2 shall not be wrap-tested unless otherwise specified in the contract or order.

3.5 Dimensional.

3.5.1 Wire.

3.5.1.1 Diameter. The permissible tolerances in diameter of all conditions of wire shall be as shown in table III.

3.5.1.2 Out-of-roundness. Wire shall not be out-of-roundness by more than one-half the total permissible tolerances shown in table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:54Z
Check the source to verify that this is the current version before use.

Table III

TOLERANCES IN DIAMETER OF COLD-DRAWN WIRE, GRADES A AND B

Specified diameter (inch)	Tolerances (inch)
To 0.0044, inclusive	±0.0002
Over 0.0044 to 0.0079, inclusive	±0.00025
Over 0.0079 to 0.0149, inclusive	±0.0003
Over 0.0149 to 0.0199, inclusive	±0.0004
Over 0.0199 to 0.031, inclusive	±0.0005
Over 0.031 to 0.045, inclusive	±0.0006
Over 0.045 to 0.079, inclusive	±0.0007
Over 0.079 to 0.1875, inclusive	±0.001
Over 0.1875 to 0.312, inclusive	+ 0.001, -0.002

4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Lot.

4.1.1.1 For visual inspection and mechanical tests. Unless otherwise specified in the contract or order, a lot shall consist of material of one grade, condition, and size offered for delivery at one time.

4.1.2 For chemical analysis. A lot shall consist of not more than 5000 pounds of material from one melt offered for delivery at one time. Where material cannot be identified by melt, a lot shall consist of not more than 500 pounds of material of one grade, condition, and size offered for delivery at one time.

Source: https://assist.dla.mil -- Downloaded: 202610-18T12:54Z
Check the source to verify that this is the current version before use.

4.2 Sampling for Government analysis. Samples shall be obtained from each lot as defined in 4.1.2. The samples shall be selected in a manner calculated to disclose any nonuniformity of material within the lot. The samples selected shall consist of not less than 2 ounces of chips or wire clippings. For fine size wires 0.032 inch in diameter and less, samples for Government analysis may be taken from the final starting size prior to the final reduction to finished size. The samples shall be forwarded to a Government laboratory designated by the procuring activity.

4.3 Visual and dimensional inspection.

4.3.1 Sampling. From each lot as defined in 4.1.1, representative samples shall be selected for inspection of visual and dimensional characteristics with sample size and acceptance criteria determined in accordance with MIL-STD-105, inspection level II, AQL 1.0 percent, unless otherwise specified by the procuring activity.

4.3.2 Each sample selected in accordance with 4.3.1 shall be subjected to surface inspection for workmanship and dimensions. Any sample unit containing one or more visual or dimensional defects shall be rejected, and if the number of defective units in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. A rejected lot may be resubmitted for Government inspection provided the manufacturer, after having been informed of the reasons for rejection, has inspected each unit in the lot for the deficiency noted and has removed, reworked, or repaired all nonconforming units.

4.3.3 Macroscopic. Macroscopic tests shall be made in full sections of the wire in the as-received condition. Tests shall be performed on samples not less than 3 inches long, taken from each end of each coil of wire. The ends of the sample shall be square-cut and fine-ground or polished. The etching solution shall be aqua regia composed of one part, by volume, of nitric acid (specific gravity 1.42) and 3 parts, by volume of hydrochloric acid (specific gravity 1.19). Etch at 95° to 100° F for 5 to 10 minutes. Inspection by a 10X magnifier will be permitted.

4.4 Sampling for mechanical tests. Samples for mechanical tests shall be selected in accordance with table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:54Z
Check the source to verify that this is the current version before use.

Table IV

SAMPLING FOR MECHANICAL TESTS

Lot size	Tension tests		Macro and wrap tests	
	No. of samples	Maximum No. of failures allowed	No. of samples	Maximum No. of failures allowed
Where material can be identified by melt				
Sample size and acceptance criteria shall be determined in accordance with MIL-STD-105, inspection level L-4, AQL 4.0 percent. Lot sizes shall be based on the number of pounds per lot.				
Condition 2 wire				
5000 lb and less	1	0	-	-
Where material cannot be identified by melt				
Condition 1 wire				
Individual coils	1	0	1	0
Condition 2 wire				
Individual coils	1	0	-	-

4.4.1 Tension, wrap, and macroscopic tests.

4.4.1.1 Where material can be identified by melt. Unless otherwise specified in the contract or order, tension, wrap, and macroscopic specimens shall be selected as indicated in table IV.

4.4.1.2 Where material cannot be identified by melt. One tension, one wrap, and one macroscopic specimen shall be selected from individual coils of wire as indicated in table IV.

4.4.2 Response to age-hardening treatment. One tension test specimen shall be selected from each coil of condition 1 wire in a lot as defined in table IV for proof of ability to respond to ageing heat treatment in agreement with the requirements of table II. The ageing temperatures are those noted in 6.3.

Source: https://assist.dla.mil -- Downloaded: 202640-18T12:54Z
Check the source to verify that this is the currentversion before use.

4.4.3 Rejection. If any sample fails to comply with the associated acceptance criteria, the lot represented shall be rejected. Rejected lots may be permitted a maximum of one resubmission subsequent to the manufacturer's removal or reworking of all defective material in the lot.

4.5 Mechanical tests.

4.5.1 Tension. Tension test specimens shall be of the full cross section of wire and not less than 15 inches in length. Specimens shall be free from sharp bends or kinks. The distance between the jaws of the testing machine with the specimen in place ready for testing shall be not less than 10 inches.

4.5.2 Wrap. Wrap test specimens shall be full-size sections of wire of a suitable length. The specimens shall be wrapped around a rod, as indicated in 3.4, and examined for cracks.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A or C as specified (see 6.2).

5.1.1 Level A. The subject commodity shall be packaged, packed, and identified, in the same manner as specified for comparable stainless steel products, in accordance with MIL-STD-163, except that no protective coatings are required.

5.1.2 Level C. The subject commodity shall be prepared for shipment in accordance with commercial practice to insure carrier acceptance and safe delivery at destination and shall meet, as a minimum, the requirements of carrier rules and regulations applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129. Nomenclature shall be the exact nomenclature for the part or items that have been approved under the contract or order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:54Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use.

6.1.1 Nickel-base alloy, grades A and B, are used for the same mechanical purposes; grade B material has special qualities for electrical service.

6.1.2 Nickel-base alloy wire, grades A and B, condition 1, is intended for coiling of helical springs which are to be age-hardened after coiling in order to obtain maximum strength and hardness. Where helical springs are to be cold pressed, the cold pressing shall be done after coiling and ageing.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Grade and condition

c. Size required (see applicable table)

d. Whether wrap-testing of condition 2 wire is required (see 3.4.2)

e. If tolerances other than those shown are required (see 3.5)

f. Lot definition, if other than 4.1.1

g. Laboratory to receive samples for Government analysis (see 4.2)

h. Sampling for visual and dimensional inspection, if other than 4.3.1

i. Level of packing required (see 5.1).

6.3 Age-hardening treatment.

6.3.1 Grade A (Duranickel). Age-harden 8 hours at 980°± 15° F in sulfur free dry hydrogen or dissociated ammonia. Cool in furnace for 6 hours minimum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:54Z
Check the source to verify that this is the current version before use.

6.3.2 Grade B (Permanickel). Age-harden 8 hours at 910 ± 15° F in sulfur free dry hydrogen or dissociated ammonia. Cool in air.

6.4 Availability. Nickel-base alloy wire is commercially available in the conditions and in the nominal ranges and increments of sizes as shown in table I. The data given in this table are informative only and subject to change. The manufacturer should be consulted for latest information available.

Custodian:
Navy - OS

Preparing activity:
Navy - OS
(Project No. 9525-N004)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:54Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-W-19487A(OS) Wire, Nickel-Base Alloy, Age Hardenable

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:54Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commanding Officer
Naval Ordnance Station
Attn: Standardization Division (502)
Indian Head, MD 20640

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:54Z
Check the source to verify that this is the current version before use.